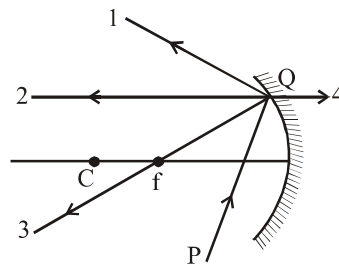


RAY OPTICS

1. The phenomenon of scattering obey Rayleigh's law i.e. $I \propto \frac{1}{\lambda^4}$. Therefore, then the traffic signals are of red colour. In this case the scattering particle (air molecules) should have size
 - (1) greater than the wave length
 - (2) smaller than the wave length
 - (3) of the order of wavelength
 - (4) None
2. Paraxial rays in geometrical optics are those rays which are
 - (1) parallel to principal axis making large angle of incidence
 - (2) non-parallel to principal axis
 - (3) parallel to principal axis making small angle of incidence
 - (4) None
3. If an object is placed in front of a concave mirror and the lower half of the mirror is covered with opaque material then what happens
 - (1) lower half of the object will be seen as image
 - (2) upper half of the object will be seen as image
 - (3) No change in the image will taken place
 - (4) image will be completely formed with less intensity
4. Optical fibre consists of core and cladding whose refractive index are related as
 - (1) $\mu_{\text{core}} = \mu_{\text{cladding}}$
 - (2) $\mu_{\text{core}} > \mu_{\text{cladding}}$
 - (3) $\mu_{\text{core}} < \mu_{\text{cladding}}$
 - (4) No relation between them
5. Dispersion occurs due to
 - (1) Refractive index of medium for different frequencies is same
 - (2) Refractive index of medium for different frequencies is different
 - (3) Refractive index is different for same frequency
 - (4) It does not depends on frequency
6. Sun is visible a little before sunrise and until a little after the actual sunset due to
 - (1) Reflection
 - (2) Refraction
 - (3) Scattering
 - (4) Total internal reflection
7. While moving in a bus or a car during a hot summer day, a distant patch on road appears to be wet. It is due to
 - (1) Reflection
 - (2) Total internal reflection
 - (3) Scattering
 - (4) Dispersion
8. Clouds which have droplets of water appear to white. This happens because of
 - (1) Dispersion
 - (2) Scattering
 - (3) Chromatic aberration
 - (4) Total internal reflection
9. In modern microscopes, multi component lenses are used for both objective and eye lens to improve image quality by minimising.
 - (1) Reflection
 - (2) Optical aberration
 - (3) Magnifying power
 - (4) Scattering
10. Thick lenses shows chromatic aberration due to
 - (1) Refraction
 - (2) Total internal reflection
 - (3) Dispersion
 - (4) Scattering
11. Rainbow formation is due to
 - (a) Refraction
 - (b) Dispersion
 - (c) Internal reflection
 - (d) Scattering
 - (1) a and b
 - (2) a, b and c
 - (3) b and c
 - (4) b, c and d
12. The condition for observing a rainbow are tha sun should be shining in one part of the sky while it would be raining in
 - (1) same part of sky
 - (2) opposite part of sky
 - (3) rain is not required
 - (4) it can rain anywhere
13. In secondary rainbow there are
 - (1) 1 internal reflection
 - (2) 2 internal reflection
 - (3) No internal reflection
 - (4) Depends on the size of water droplet

14. Modern telescopes are designed in such a way that there is no chromatic aberration present. It is used by having
- (1) Convex lens
 - (2) Concave mirror
 - (3) Both concave mirror and convex lens
 - (4) Prism
15. Laws of reflection are strictly valid for
- (1) Plane surfaces
 - (2) Rough surfaces
 - (3) All types of surfaces
 - (4) None of these
16. A short pulse of white light is incident from air to a glass slab at normal incidence. After travelling through the slab, the first colour to emerge is
- (1) Blue
 - (2) Green
 - (3) Violet
 - (4) Red
17. A passenger in an aeroplane shall
- (1) Never see a rainbow
 - (2) may see a primary and a secondary rainbow as concentric circles
 - (3) may see a primary and a secondary rainbow as concentric arcs.
 - (4) shall never see a secondary rainbow
18. In telescope, which mirror is used as objective instead of convex lens
- (1) Parabolic concave mirror
 - (2) Plane mirror
 - (3) Convex mirror
 - (4) Mirror can't be used
19. Objective lens of telescope has large aperture
- (1) to increase intensity and resolving power
 - (2) to reduce resolving power
 - (3) to obtain small magnifying power
 - (4) to reduce intensity
20. In vacuum, all colours
- (1) have same speed
 - (2) have different-different speed
 - (3) do not move
 - (4) absorb all colours
21. A person can see his inverted image in a concave mirror when he is
- (1) between focus and center of curvature (C)
 - (2) beyond C
 - (3) between focus & pole
 - (4) at focus
22. Which of the following colours of white light deviated most when passes through a prism.
- (1) Red light
 - (2) Violet light
 - (3) Yellow light
 - (4) Both (1) & (2)
23. An under-water swimmer cannot see very clearly even in absolutely clear water because of
- (1) absorption of light in water
 - (2) scattering of light in water
 - (3) reduction of speed of light in water
 - (4) change in the focal length of eye lens
24. The direction of ray of light incident on a concave mirror is shown by PQ while direction in which the ray would travel after reflection is shown by four rays, marked 1,2,3 and 4 which of the four rays correctly shows the direction of reflected ray ?

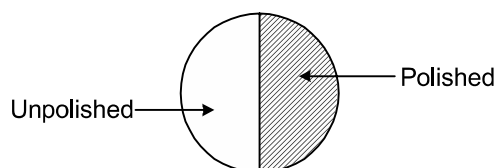


- (1) 1
 - (2) 2
 - (3) 3
 - (4) 4
25. Optical denseness of a medium is measured in terms of ____.
- (1) refractive index
 - (2) mass density
 - (3) (1) & (2) both
 - (4) can't measure
26. An air bubble is formed inside water. It act as a :-
- (1) convex mirror
 - (2) converging lens
 - (3) diverging lens
 - (4) plane mirror

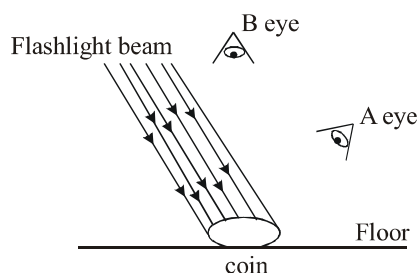
27. An object is first seen in red light and then in violet light through a simple microscope. In which case, the magnifying power is more.
 (1) violet light
 (2) red light
 (3) same in both light
 (4) can't see magnified image
28. A prism is made up of flint glass whose dispersive power is 0.053. Find the angle of dispersion if the mean refractive index of flint glass is 1.68 and the refracting angle of prism is 3° .
 (1) 20.08° (2) 10.08°
 (3) 0.208° (4) 0.108°
29. How does refractive index (μ) of a material vary with respect to wavelength (λ) ? A and B are constants

- (1) $\mu = A + B\lambda^2$ (2) $\mu = A + \frac{B}{\lambda^2}$
 (3) $\mu = A + B\lambda$ (4) $\mu = A + \frac{B}{\lambda}$

30. Consider the metal coin in figure. The right half is polished, while left half is unpolished.



In a dark room, this coin is placed face up on the floor and illuminated with a thin-beamed flashlight, as shown in figure. A and B look at the coin from the position as indicated in figure.



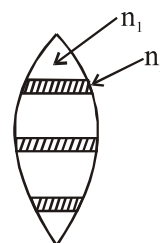
According to A, which half of the coin (if either) is brighter?

- (1) The polished half
 (2) The unpolished half
 (3) Both halves look equally bright
 (4) Both halves look completely dark

31. A diminished image of an object is to be obtained on a screen 1.0 m away from it. This can be achieved by approximately placing:

- (1) a convex mirror of suitable focal length
 (2) a concave mirror of suitable focal length
 (3) a convex lens of focal length more than 0.25 m
 (4) a concave lens of suitable focal length.

32. Shown in the figure is a bi-convex lens made of different transparent strips of two different refractive indexes. If the parallel rays of light is incident on the lens then select INCORRECT statement



- (1) rays will intersect before focussing
 (2) rays will be focussed at two different points
 (3) the lens may have four focal points different from each other.
 (4) None of the these
33. Between the primary and secondary rainbows, there is a dark band known as Alexander's dark band. This is because :-
 (1) Light scattered into this region interfere destructively
 (2) There is no light scattered into this region
 (3) Light is absorbed in this region
 (4) Angle made at the eye by the scattered rays with respect to the incident light of the sun lies between approximately 42° and 50°
34. A magnifying lens is used, as the object to be viewed can be brought closer to the eye than the normal near point. This :-
 (1) A larger angle to be subtended by the object at the eye and hence, viewed in greater detail.
 (2) The formation of real erect image.
 (3) Increase in the field of view
 (4) Infinite magnification at the near point

35. The phenomena involved in the reflection of radiowaves by ionosphere is similar to :-

- (1) Reflection of light by a plane mirror
- (2) Total internal reflection of light in air during a mirage
- (3) Dispersion of light by water molecules during the formation of a rainbow
- (4) Scattering of light by the particles of air

36. The apparent depth of a tank filled with water:-

- (1) Will be more for oblique viewing
- (2) Will be more for normal viewing
- (3) Is independent of the angle of viewing
- (4) None of above